Course: **Data Structure** **Semester: 2st term 2025/2026.**	**Lecturers: Dr. Eman Shawky** **TAs: Eng.Mostafa Elmansy, Eng.Menna, Eng.Ragab, Eng.Heba, Eng.Lamiaa, Eng.Rana**
Sheet 1	

MCQ – Data Structure & Difference Between Data Structure and Database

1. What is a Data Structure?

A) A programming language
B) A way to store and organize data
C) A type of database
D) An operating system

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2. Which of the following is a linear data structure?

A) Tree
B) Graph
C) Array
D) Heap

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3. What is the main purpose of a Data Structure?

A) Store large data permanently
B) Organize data for efficient processing in memory
C) Manage users
D) Store backup files

4. What is the main purpose of a Database?

A) Execute algorithms
B) Store and manage large data permanently
C) Replace programming languages
D) Design user interfaces

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5. Which one supports queries like search, update, delete using query languages?

A) Data Structure
B) Database
C) Stack
D) Array

Question 1

Write a C++ program that reads n elements from the user into an array, then displays the elements in the same order they were entered.

Question 2

Write a C++ program to determine and display the maximum element stored in an array.

Question 3

Write a C++ program that reads n elements into an array, then uses a separate function to calculate the sum of the array elements, and finally displays the result.

Question 4

Write a C++ function that calculates and returns the average value of elements stored in an array initialized with the values {10, 20, 30, 40, 50}.

Question 5

Write a C++ program that uses an array initialized with the values {1, 2, 3, 2, 1}, then uses a function to determine whether the array is a palindrome (same forward and backward), and prints the result.